

TRUE PHYSICS

(unknown to the estimator)

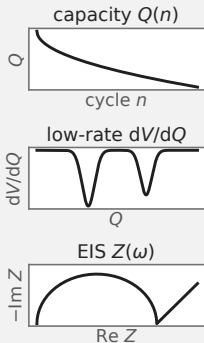
SPE, Eqs (1)–(4): solid diffusion,
electrolyte transport, Butler-Volmer
→ voltage $V(t)$

- SEI growth
- lithium plating
- particle cracking
- loss of active material

rate constants k_{SEI} , $j_{0,\text{plate}}$, β_{LAM} , k_{cr}
per-mechanism loss $Q_m(n)$, Eq. (5),
used only to score

observe

WHAT YOU MEASURE



attribute

$\hat{m}$, $\hat{f}_m$

WHAT SURVIVES



LAM: fraction partial
3/6 label; fraction $R^2 = +0.53$



SEI: not recovered
2/7 label; fraction $R^2 = -0.77$



plating: never identified
0/5 label; fraction $R^2 = -0.77$



cracking: label only
4/6, an artefact of near-zero fade

NOT recovered: the governing equations · the rate constants · the internal state

Only a dominant-mechanism label and the mixing fractions are recovered, scored against the hidden decomposition.